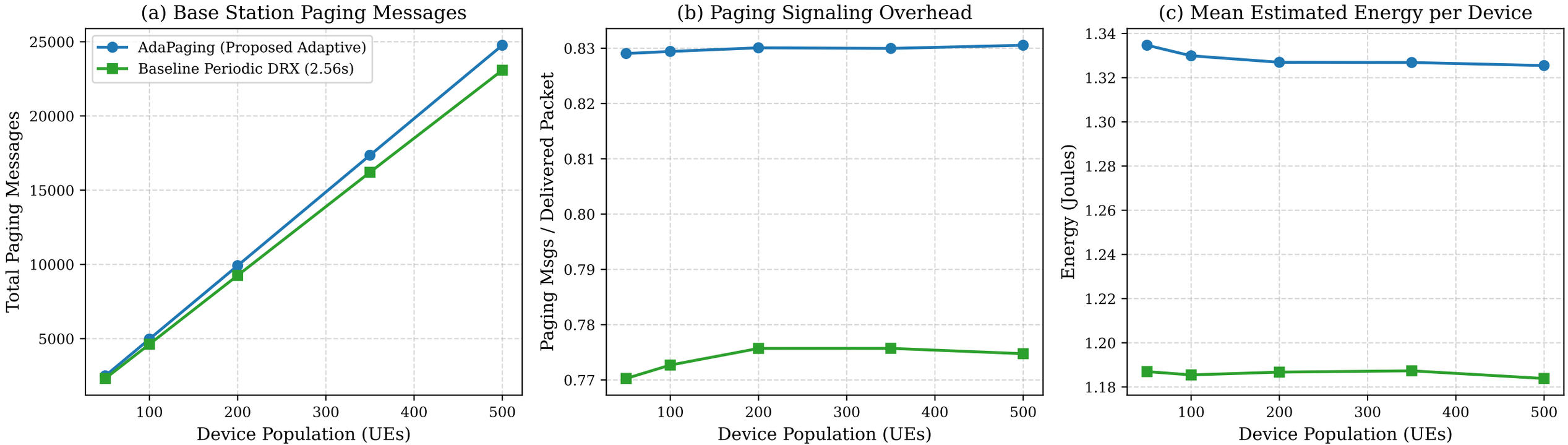
Scalability Analysis Across Device Population
(a) Base Station Paging Messages
AdaPaging (Proposed Adaptive)
Baseline Periodic DRX (2.56s)
Total Paging Messages
25000
20000
15000
10000
5000
100
200
300
400
500
Device Population (UEs)
(b) Paging Signaling Overhead
Paging Msgs / Delivered Packet
0.83
0.82
0.81
0.80
0.79
0.78
0.77
100
200
300
400
500
Device Population (UEs)
(c) Mean Estimated Energy per Device
Energy (Joules)
1.34
1.32
1.30
1.28
1.26
1.24
1.22
1.20
1.18
100
200
300
400
500
Device Population (UEs)